# UNIT 2: SHAPES AND REGIONS

## SESSION 9: REGION DESCRIPTORS

## SLO 1: DESCRIBE VARIOUS REGION-BASED DESCRIPTORS USED FOR CHARACTERIZING OBJECT SHAPES IN IMAGES.

**Pick the Odd One Out**

**From the given options, identify the odd one out and briefly explain why.**

1. Area – Centroid – Perimeter – Orientation

   Odd one: Perimeter

   Reason: Perimeter is a boundary-based descriptor, while area, centroid, and orientation are region-based descriptors.

2. Area – Eccentricity – Compactness – Brightness

   Odd one: Brightness

   Reason: Brightness describes intensity, while area, eccentricity, and compactness describe the shape or region.

3. Centroid – Orientation – Texture – Area

   Odd one: Texture

   Reason: Texture describes surface patterns or intensity variations, while centroid, orientation, and area are region descriptors.

4. Region Descriptor – Boundary Descriptor – Shape Descriptor – Color Descriptor

   Odd one: Color Descriptor

   Reason: Color descriptor describes color information, while the others are related to object shape or region description.

**Descriptor Sorting Challenge**

**Below are different image features.**

**Classify them as Region Descriptor or Not a Region Descriptor.**

| Feature | Region Descriptor / Not |
|---|---|
| Area | **Region Descriptor** |
| Centroid | **Region Descriptor** |
| Orientation | **Region Descriptor** |
| Edge gradient | **Not** |
| Eccentricity | **Region Descriptor** |